# CL-TAV: Closed-Loop Test–Analysis Verification for ARINC 615A Data Loading under Timing Uncertainty

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***Abstract*— Airborne software loading is judged from protocol events whose identity, ordering and timing may be only conservatively observed. This paper specifies Closed-Loop Test–Analysis Verification (CL-TAV) for ARINC 615A: a charged session that predicts observation classes, interprets a raw record under one uncertainty reference, updates a finite single-fault candidate set from the whole compatible history, and selects one exclusive TEST, Prep, Recover or Admit action. The contribution sought is deliberately bounded. C1 is consistency and conditional true-hypothesis retention under the declared model, ownership and conservative-observation assumptions, not a new general theorem for loss, drift or reordering. C2 is a prior-free one-step minimax choice over currently feasible and affordable tests with explicit error and stop semantics, not a globally optimal policy. The bound protocol model remains UPLOAD and INFORMATION; ARINC 645 identity is bound for triggered leaves, while runtime integrity remains unestablished. Four comparison arms and three experiment families are registered. Confirmatory measurements are not yet available, so no improvement, implementation-completeness or certification claim is made.**

***Index Terms*— Avionics, data loading, conformance testing, fault diagnosis, timing uncertainty, closed-loop verification.**

## I. Introduction

Airborne software loaders exchange INFORMATION and UPLOAD traffic under declared timers, file identities and operator actions [1]. Loading frameworks also treat identity, integrity, interrupted loading and version conflict as connected engineering concerns [2]. A missed or late acknowledgement, a mismatched Load File, or an unresolved integrity check can therefore be product behaviour, a capture or clock limitation, a configuration discrepancy, or an unconfirmed injection. The engineering problem is not merely to collect more tests, but to decide whether the current evidence is usable and which later protocol-legal action is worth its finite resource.

This paper studies Closed-Loop Test–Analysis Verification (CL-TAV). Three reader questions are used throughout:

- CL-RQ1 (loop consistency): do prediction, interpretation and candidate update stay mutually consistent under one uncertainty reference?
- CL-RQ2 (comparative effect): under the same budget and denominators, does the closed loop change detection or isolation relative to Test-only, Analysis-only and fixed Test-then-Analysis arms?
- CL-RQ3 (uncertainty and stopping): how do interval timing, `ERROR`, Prep/Recover and exclusive stops affect those comparisons?

Two propositions are evaluated, not asserted as established results. C1 is mutual consistency of uncertainty-aware interpretation, timing verdict and whole-history compatible update in the declared finite domain; its retention statement is conditional on model, ownership, pairing and conservative observation assumptions. C2 feeds residual ambiguity into a prior-free, one-step worst-case candidate-set choice among feasible affordable tests, with explicit Prep/Recover, `ERROR`, resource and stop rules. C2 is not a global multi-step optimum and need not shrink the candidate set on every outcome.

Requirements-based generation, conformance testing and mutation adequacy supply important test-construction and evaluation tools [3]–[6]. Active diagnosis already closes observation–control loops, while real-time diagnosis and testing already address uncertain communication timing [7]–[11]. The narrower question here is whether the declared 615A action, observation, cost and stop contracts form a useful and empirically defensible combination. Section II therefore compares stated objects and assumptions; it does not infer novelty from different terminology or prove that another method cannot be extended.

The evaluation uses four arms (CL-T, CL-A, CL-TA, CL-LOOP) and three registered families (DETECT, LOCATE, ABLATION). Confirmatory numbers are not available.

**Placeholder PH-ABS-RESULT**
(EXPERIMENT-RESULT)

Abstract and conclusion improvement statements remain empty until registered runs exist.

Data are not obtained. This box reserves layout. It is not a result.

Section II defines the objects. Section III separates CRS layers and the two machines. Section IV gives the loop, update and T5 mathematics. Section V freezes the comparison contract. Sections VI–VIII reserve results and limits. The separately compiled supplement typesets wrappers ALG-CLTAV-02/03/04, core mechanisms 05/06/07, typed records and the A/P decision tables. Paper Algorithm numbers are display labels; stable ALG-IDs are unchanged.

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## II. Related Work and Problem Formulation

### A. Evidence-bounded comparison

Table I organizes the closest evidence by problem strand rather than by paper. It records the compared object and the boundary that prevents a category-level novelty claim.

TABLE I
Closest evidence strands and the resulting CL-TAV boundary.

| Strand | Existing object | Boundary used here |
|---|---|---|
| Uncertain history | compatible-run diagnosis under latency, drift/offset and reordered observation; exact and bounded-history variants [10]; delay-aware monitoring/testing [11] | C1 is conditional project consistency, not the invention of uncertainty-aware history diagnosis |
| Active choice | closed-loop supervisors, conditional plans and time-interval control [7]–[9], [12] | C2 is one-step residual-set selection with a different action, cost and stop contract, not the first active loop |
| Aviation loading | onboard diagnostic procedures and avionics MBT [13]–[15]; proposed LAS requirements/procedures [2] | engineering context, not executed 615A conformance evidence |
| Evaluation | mutation denominators and equivalent/undecided mutants [5], [16]; randomized-algorithm reporting [17] | informs accounting and registration, not CL-TAV performance |

Köhl and Hermanns are the strongest C1 neighbour found. Their passive diagnoser ranges over runs compatible with the whole timed observation history under bounded action-specific latency and a clock remapping, including latency-induced reordering [10]. Their exact incremental algorithm is separated from a sound but generally incomplete fixed-history approximation. CL-TAV therefore does not claim history compatibility or uncertainty handling as new, does not call their "active prefixes" active tests, and does not inherit exact-once observation or their liveness assumption. Fränzle *et al.* likewise give exact delay/jitter-aware monitoring and testing semantics, but testing there is harness input plus verdict evaluation rather than adaptive next-action optimization [11].

Active diagnosis has long combined online state estimation with controlled behaviour [7]. Conditional action plans, state-conditioned event disablement and time-interval restrictions further establish feedback and active choice as prior art [8], [9], [12]. Some objectives use costs or probabilistic information. CL-TAV instead fixes the current finite feasible set and ranks an affordable distinguishing TEST by the one-step worst residual cardinality, then cost and identifier. The lack of a default prior is a design choice, not an automatic advantage; Prep/Recover and exclusive error/stop semantics are compared as engineering contract differences, not proof of first combination or global optimality.

For the application bridge, onboard active diagnosis demonstrates procedure generation and also reports wrong localization under limited observability and model/prototype mismatch [13]. IMA and cockpit-display work demonstrate configuration-aware, model-derived avionics testing [14], [15]. F-LAS proposes 23 reusable requirements and six procedures across normal, interrupted/incomplete and version-conflict scenarios, but reports neither procedure execution nor an ARINC 615A conformance result [2]. These sources motivate cases and threats; the standard and the controlled CRS remain the protocol authority. All comparisons above are bound to the full-text versions recorded in the evidence ledger; no absence claim is inferred from an unreported feature.

### B. Verification objects

The IUT is a 615A data-loading endpoint. The bound protocol model is UPLOAD and INFORMATION [1]. FIND, DOWNLOAD and conditional AFDX appear in the CRS; they are not implemented behaviour and not an experimental arm.

Candidates are $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$. An observable summary $q$ is known or unknown. Observation classes are projected onto current $H$. Measurement uncertainty is a declared conservative outer set $U$ in the GUM sense [18], [19], not an estimated physical error model. Resource is exactly one mode: BUDGET$(c_{\min}, B)$ or ROUNDS$(K_{\max})$.

A 615A-triggered 645 bind covers CRC, check-value and naming leaves [20]. Integrity capabilities remain unestablished. Stop-645 names a remaining 645-dependent obligation, not a missing file.

## III. System Engineering and Verification Architecture

Protocol CRS, tool requirements and experiment requirements stay in separate layers. Row count does not prove a method. A concrete owned trace is source unit `SU-ARINC-615A-3-SECTION-6-3-2-SEQUENCE-CHART-A-E007-7726A816FFE7` → CRS-M1-00365/00366/00367 → LUR write-order constraint → planned Test/Analysis verification case → later tool module. The IDs are real bound-M1 identifiers. `satisfy/verify` in the SysML view are model relations, not executed verification. No standard figure is reprinted.

The layering responds to two aviation lessons without importing their authority. F-LAS shows the usefulness of explicit requirements-to-procedure traceability for loading identity, integrity, interrupted loading and version conflict, but its procedures were proposed rather than executed [2]. Avionics MBT also makes configuration, model preparation, simulator and oracle boundaries part of the test problem [14], [15]. Accordingly, the CRS constrains legal protocol evidence, the tool realizes method contracts, and

the evaluator owns independent truth; none can silently stand in for another.

Two machines must not be merged. $M_{\text{sess}}$ selects from $S \subseteq A(q)$, charges once, updates $H$ and returns a named stop. $M_{\text{prot}}$ is the UPLOAD/INFORMATION black box. Execute issues a stimulus and obtains a raw execution record. Interpretation, not the network port, produces the compatible class $I_z$ and hypothesis survivors.

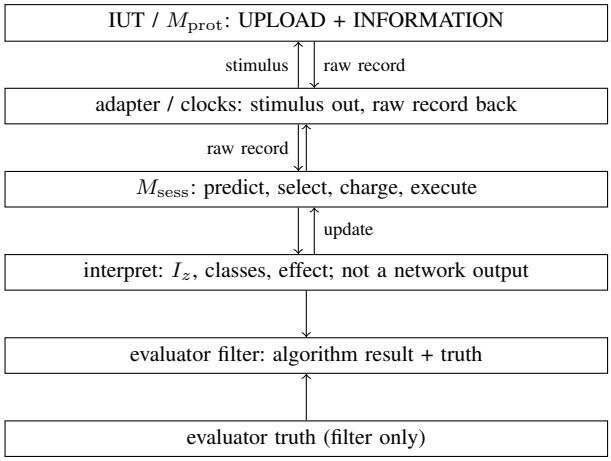



Fig. 1. Context derived from FIG-CL-TAV-01/07. Stimulus and raw records share ports. Truth enters only the evaluator filter, never select, predict, history or execute.

Expanded CRS leaves do not add FIND or DOWNLOAD transitions to $M_{\text{prot}}$. The development contract now binds eight algorithm interfaces and seven experiment interfaces to typed inputs, outputs, ownership, failure, resource, acceptance, and successor implementation locations. The first slice also preserves six CRC-bearing protocol obligations and their ARINC 645 parameter leaves. The six contracts remain distinct: INFORMATION and UPLOAD receiver support, final-image/LSP check value, optional old/new-file comparison, equal CRC values for two distinct files sharing one part number, and status-file continuation after the final DATA while integrity calculation remains in progress. This closes their specification dependency only: the runtime integrity capability remains *not established* until implementation, project configuration, controlled-vector execution, and an independent gate record exist. Missing evidence therefore yields `NOT-EVALUATED`, not a protocol verdict.

## IV. CL-TAV Method

### A. Notation and assumptions

S0–S10 are logical duties, not line numbers. The decision index $k$, the billed charge count and the retry counter are distinct. SessionContext $\Gamma$ is the unique session state. HistoryHandle $\eta$ carries whole-history compatibility. The select-time snapshot $\xi$ freezes `actionId`, `actionKind`, $q_{\text{used}}$, history version, classes used at select, and uncertainty reference $U$. A later confirmed successor summary is a different field. Effect classes are `CONFIRMED-NOT-SENT` and `UNKNOWN-EFFECT`. PredictionGapError is TEST-scoped.

### B. Overall closed loop

Algorithm 1 is a typed abstraction of ALG-CLTAV-01. Wrappers PREDICTCURRENT, INTERPRETOUTCOME and RESOLVEOUTCOME are ALG-CLTAV-02/03/04 in the supplement. S9 calls COMMITCOMPATIBLEUPDATE (ALG-CLTAV-07), with $\chi = (\Gamma, \eta, H)$, which validates the history version, applies `IF-HIST-UPDATE` from $\xi$, returns $(\textit{status}, \eta, H, \Gamma)$ and commits a confirmed successor summary. A bare `IF-HIST-UPDATE` call is not that wrapper. Resolution is branched on $\delta.\text{control}$ after adopting $\delta.\Gamma$, $\delta.\eta$ and $\delta.\text{outcome}$.

### C. Interpretation, compatibility and selection

Prediction and interpretation share $U$. Let $O(h, t, q)$ be the conservative outer set of feasible observations of hypothesis $h$, and let $Z_k$ be the measured feasible-observation set after timing, ownership and pairing checks. The interpretation/whole-history bridge induces the compatible hypothesis class $C_k$ (the accepted interface field $z.I_z$); $C_k$ is not a measurement set. A currently valid class is the nonempty projection $H \cap C_k$. Compatible update on a valid class is

$$H_{k+1} = H_k \cap C_k, \qquad H_{k+1} \subseteq H_k. \tag{1}$$

`ERROR` and identity Prep/Recover leave $H$ unchanged. The one-step score on currently valid nonempty classes of an affordable distinguishing TEST $t$ is

$$s(t) = \max_{C \in \mathcal{C}_t} |C|. \tag{2}$$

Empty $S$ is never $\arg\min$. Timing uses T5 [19]: $I_{\text{obs}} \subseteq I_r$ is PASS; $I_{\text{obs}} \cap I_r = \varnothing$ is FAIL; otherwise INCONCLUSIVE. An ambiguous pair is `ERROR`, not IUT FAIL.

### D. Bounded reference-kernel contract

The first implementation target is a finite explicit-path kernel, not a general solver. Each frontier entry contains a stable hypothesis ID, control state, canonical exact-rational difference constraints, finite typed store, path length, whole-history provenance, and a known/conservative-unknown status. It accepts only TRUE, conjunction, state equality, typed-field equality and rational interval guards, with simultaneous finite assignment and exact zero-clock-reset updates. OR, NOT and calls are outside this first domain and are rejected as SPEC-ERROR before expansion; no normalization is left to an implementer. The sole normative numeric form is a numerator and strictly positive denominator pair; interval bounds carry explicit open/closed flags, and any fraction printed in a review view is display-only. A typed adapter either maps the supported M2 subset into this AST or rejects the model before expansion, so M2 expressions are not a second authoritative encoding. Propagation validates

**Algorithm 1:** CL-TAV top-level session (paper Alg. 1 = ALG-CLTAV-01)

**Require:** $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$; $\Gamma$; $\eta$; one mode BUDGET$(c_{\min}, B)$ xor ROUNDS$(K_{\max})$; $L$; $U$
**Ensure:** stop class, final $H$, $\eta$, trace; singleton $h_{\text{normal}}$ is not protocol PASS

$(\Gamma, \eta, H) \leftarrow$ INITIALIZE$(H_0, L, U, \text{resourcePolicy})$;
**while** *true* **do**
  *stop* $\leftarrow$ IF-RESOURCE-STOP$(\Gamma, H, \eta)$ with IF-EQUIV$(\Gamma, H, \eta)$;
  **if** *stop* $\neq$ *undecided* **then return** FINISH$(stop, \Gamma, \eta, H)$;
  *pred* $\leftarrow$ PREDICTCURRENT$(\Gamma, \eta, H, U, L)$;
  *dec* $\leftarrow$ SELECTANDADMIT$(\Gamma, \eta, H, L, pred)$;
  **if** *dec*.kind $\in \{EXIT, SPEC\text{-}ERROR\}$ **then return** FINISH$(dec.\text{reason}, \Gamma, \eta, H)$;
  $\xi \leftarrow$ SELECTSNAPSHOT$(dec; \Gamma, \eta, U)$;
  $\xi = (\text{id}, \text{kind}, q_{\text{used}}, v_\eta, C, U)$;
  kind $= \xi$.actionKind;
  $C = \xi$.classesUsedAtSelect;
  $\Gamma \leftarrow$ CHARGEONCE$(\Gamma, \xi)$; *er* $\leftarrow$ IF-EXECUTE-RECORD$(\Gamma, \xi)$;
  $z \leftarrow$ INTERPRETOUTCOME$(er, \xi, \Gamma, \eta, U)$;
  $\delta \leftarrow$ RESOLVEOUTCOME$(z, \xi, \Gamma, \eta)$;
  $\Gamma \leftarrow \delta.\Gamma$; $\eta \leftarrow \delta.\eta$; $z \leftarrow \delta$.outcome;
  **if** $\delta$.control $= STOP$ **then return** FINISH$(\delta.\text{reason}, \Gamma, \eta, H)$;
  **if** $\delta$.control $= RETRY$ **then continue** to S1 with adopted $\Gamma, \eta$;
  **if** $z.I_z$ *valid or* $z$.summaryConfirmed **then**
    *upd* $\leftarrow$ COMMITCOMPATIBLEUPDATE$(\xi, z; \chi)$;
    $(status, \eta, H, \Gamma) \leftarrow upd$;
    **if** *status* $= SPEC\text{-}ERROR$ **then return** Stop-SpecError;
  **end**
  *stop* $\leftarrow$ IF-RESOURCE-STOP$(\Gamma, H, \eta)$ with IF-EQUIV$(\Gamma, H, \eta)$;
  **if** *stop* $\neq$ *undecided* **then return** FINISH$(stop, \Gamma, \eta, H)$;
**end**

the select-time snapshot, expands enabled transitions, intersects observation and timing constraints, and retains every feasible successor. Entries merge only when hypothesis, control state, clock constraint, store and history provenance are identical.

Configured bounds cover hypotheses, actions, observation classes, path length and per-hypothesis frontier size. Unsupported syntax or an exhausted path or frontier bound returns the interface's named GAP, SPEC-ERROR, or conservative unknown result without deleting a hypothesis. Only prediction, selection, history update, equivalence and stop selection consume kernel results; execute, interpretation and Prep/Recover do not. Feasibility alone never establishes equivalence. Unsupported syntax, computation unknown and limits preserve $H$, the summary and charge state, irrespective of a later stop label. A valid successor equals the registered compatibility intersection and is a subset of $H$. Its empty result forces Stop-Empty and its singleton result forces Stop-Singleton before 645, equivalence or budget exits. Stop-Empty thus requires an empty history proved by complete in-domain intersection. This contract maps every reachable result and marks every other interface/result pair unreachable; it does not claim completeness, polynomial complexity, or deployed code.

This contract is intentionally weaker than a universal uncertain-observation diagnosis theorem. Existing work already supplies possible-run semantics and an exact incremental passive diagnoser for bounded latency, drift/offset and out-of-order observation under exact-once and timed-automaton assumptions [10], as well as exact monitoring/testing for a different bounded communication-delay model [11]. CL-TAV claims only the consistency and retention property below under R1–R5. Detected capture inadequacy or invalid evidence, ambiguous ownership or pairing, unsupported syntax and exhausted bounds retain their existing named uncertain, error or resource outcomes and do not establish an IUT failure. This does not claim that every missing, duplicate or spurious event is detectable: undetected out-of-domain pollution may remain compatible with an in-domain candidate. Conversely, with qualified evidence and a valid still-active obligation, expiry at the declared timing boundary retains the existing no-response FAIL; a legal observation class whose complete in-domain intersection is empty retains Stop-Empty. No source theorem is transplanted.

## E. Conditional properties

**Monotonicity.** Equation (1) gives $H_{k+1} \subseteq H_k$. Without a valid class there is no add or delete. An in-domain singleton is not protocol PASS.

**Retention.** Assume (R1) $h^\star \in H_0$; (R2) one state path is consistent with the executed history; (R3) $O(h^\star, t_k, q_k)$ is a conservative outer set; (R4) the true observation belongs to both $O(h^\star, t_k, q_k)$ and $Z_k$; and (R5) ownership and pairing are valid. The declared interpretation and whole-history compatibility contract therefore places $h^\star$ in $C_k$. Thus $h^\star \in H_k \cap C_k$, and induction retains it in $H_{k+1}$. Out-of-domain faults may still mimic an in-domain survivor.

**One-step minimax.** On nonempty finite $T_{\text{cur}}$, $t^\star$ is a least element of $(s(t), \text{cost}(t), \text{id})$. Diagnostic-state feedback and timed active control already exist [8], [9]; this statement concerns only the declared current choice. It is not a global policy, does not prove future distinguishability, and does not guarantee shrinkage for every observation class.

**Termination.** Every charged TEST, Prep, Recover and ERROR spends the selected mode, hence at most

$\lfloor B/c_{\min} \rfloor$ or $K_{\max}$ billed executions. Wall-clock waits still need their own timeouts; those timeouts are not claimed implemented.

Exclusive named stops are Empty, Singleton, 645, Equivalent, Budget, Stop-Error and Stop-NoDistinguisher. Stop-Error is not Stop-Budget. Admit reasons A2–A5 and post-update P1–P5 are tabulated in the supplement. CONFIRMED-NOT-SENT does not assign UNKNOWN. UNKNOWN-EFFECT must not be read back as the previous KNOWN summary. Unconfirmed Recover with $R_{\max} = 1$ and remaining budget is Stop-Error. The update disposition is evaluated before the mutable stop label: a rejected/error update preserves $H$, while a valid empty or singleton update forces Stop-Empty or Stop-Singleton, respectively. Kernels remain unimplemented interfaces.

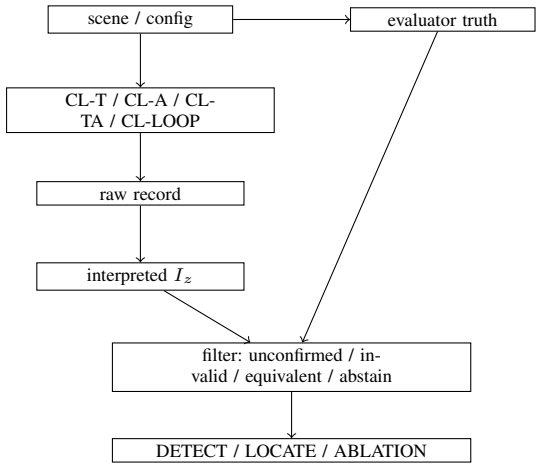



Fig. 2. Experiment architecture from FIG-CL-TAV-09. Truth meets algorithm output only at the evaluator filter.

## V. Experimental Setup and Evaluation Protocol

The comparison contract is the existing experiment plan. Four arms share the same information pool, action pool and fault domain. Direct numeric comparison is allowed only on the same denominator and the same charged projection of $c = (c_{\text{acq}}, c_{\text{comp}}, c_{\text{prep}}, c_{\text{retry}})$. Runtime records may retain recovery as a detail, but evaluation folds it into $c_{\text{prep}}$ exactly once; it is not a fifth comparison coordinate. Seven controlled interfaces define scene registration, injection, independent truth, causal collection, arm execution, validity filtering and evaluation. Historical captures use only the offline evidence line and are not runnable scenes or confirmatory truth. The controlled line still depends on an interactive IUT, confirmed injection, reset evidence, qualified clocks and a truth generator whose shared components are disclosed.

- CL-T: predetermined tests; projection $c_{\text{acq}}$ for tests that run.
- CL-A: analysis on pre-allowed records; projection $c_{\text{comp}}$ only; no invented sampling cost.
- CL-TA: frozen collect then analyse; $c_{\text{acq}}$ and $c_{\text{comp}}$ reported separately.
- CL-LOOP: the CL-TAV session; charge the components that actually run.

Truth is typed evaluator-only. It never enters `IF-SELECT-ADMIT`, `IF-PRED-OBS`, `IF-HIST-UPDATE` or `IF-EXECUTE-RECORD`; algorithm-visible references are limited to registered initial inputs and records produced by that arm's own actions. In particular, arms receive the proposed allowlisted `algorithmVisibleFields` projection, never the parent scene record containing the fault plan, truth source, injection confirmation, validity class or post-hoc adjudication. The projection is a typed member allowlist containing only scene ID, service scope, registered arm, resource mode and public-configuration reference; truth, injection, validity, equivalence and post-hoc members remain forbidden inside the object. Runtime `IF-EQUIV` candidate equivalence is distinct from evaluator-side injected-fault equivalence. The all-attempt ledger retains every registered attempt, including unconfirmed injection, invalid observation, equivalent or undecided fault, `ERROR`, abstention and unconcluded execution. Detection uses a named valid denominator and a separate answered subset for PASS/FAIL error. Localization uses confirmed in-domain non-equivalent injections and separately reports localization-decision coverage, whether the true fault remains or is excluded, whether an issued singleton is correct, and final $|H|$. These finite populations are derived from attempt fields rather than handwritten denominator tags. Missing truth remains in answered or decision coverage and an explicit truth-missing flow, but leaves only the relevant truth-scored subset. Thus a valid detection FAIL with $H = \{h_1, h_2\}$ is an answered detection outcome but not singleton localization; $H = \{h_{\text{normal}}\}$ is not protocol PASS. Mutation scores exclude only confirmed equivalents while equivalent and undecided counts and adjudication effort remain visible [5], [16]. If no attempt is answered, answered coverage is reported as $0/N$ for $N > 0$, whereas answered-conditional error is not applicable. An `ERROR`, `Stop-Empty`, `Stop-645`, or budget stop cannot manufacture an answered or singleton-localization decision. Rejected updates preserve (H) independently of the stop tag; valid empty and singleton updates force `Stop-Empty` and `Stop-Singleton` before later stops.

DETECT compares miss and refusal under the same detection budget and the T5 oracle; FIND abort does not waive the 2 s / 3 s clocks. LOCATE compares surviving $|H|$ and true-fault exclusion on confirmed in-domain injections. ABLATION uses the registered factors: no feedback (CL-T or CL-TA); Prep/Recover uncharged; interval versus point-timestamp on the same traces with clocks fixed; and FIND-abort-as-waiver as a forbidden negative control, not an allowed algorithm. Sample size remains a named gap.

**Placeholder PH-STAT-N**
(PUBLICATION-DECISION)

Numeric $n$, repeats and interval method freeze at confirmatory registration, not in this draft.

Data are not obtained. This box reserves layout. It is not a result.

The implementation-design attachment fixes the accounting contract but not run parameters. An experimental unit is a unique registered attempt on one base scene, arm and control/fault construction; packets, events, timestamps and hypotheses are within-attempt observations. Development, pilot and held-out lineages are disjoint, and paired arms require independent reset evidence. Pair eligibility is derived separately for each parent metric and additionally requires the same registered base scene, seed, construction and contrast, reciprocal distinct arm identities and distinct confirmed resets; an unmatched or mismatched row remains in flow accounting but is not imputed. Eight existing scenario IDs cover normal, WAIT/continue, no-response, abort, invalid observation, same-key ownership, single-batch comparison and resource error. Their specification coverage does not establish fault coverage. FIND, DOWNLOAD, AFDX and runtime integrity capability remain outside first-run admission. Design review therefore precedes engineering feasibility, pilot, frozen registration, execution and result acceptance.

For the no-response scenario bound to CRS-M1-00101, the scored trigger is the data loader receiving a status file that carries an exception delay for the active operation. A new status within that delay or the required data-loader abort fallback after expiry satisfies the registered timeline; absence of both fails it. Generic request/response suppression remains a method-only case and is not scored against CRS-M1-00101; the abort is a prescribed fallback, not a cancellation waiver.

The registered statistical decision must classify comparisons as primary, secondary or exploratory and map each primary estimand to its finite domain, unit, pairing, estimator, interval, applicability and effect scale. Numeric sample/repeat policy, resource budget and stop priority, and clock/error budget are separate open decisions with named owners and closure witnesses; none is selected by this draft. The finite decision table distinguishes a complete deterministic census, independently randomized attempts, paired units and clustered units. The independent branch requires independent random assignment, no dependence and attempt as the unit; the paired and clustered branches require their respective evidence and units. Missing or inconsistent inputs produce `WAIT-DEC-SAMPLE`. Only the independent Bernoulli branch currently names Clopper–Pearson from a bound direct source; paired and clustered inference remain method-wait. The examples are hypothetical, not power calculations or results. One session selects exactly one `BUDGET` or `ROUNDS` mode. A retry cap is an independent execution-safety bound, not a second budget stop. The timing witness in the design contract uses closed intervals $I_{\mathrm{obs}} = [5.0, 5.2]$ s and $I_r = [4.0, 5.1]$ s, whose overlap without containment is `INCONCLUSIVE`; containment is PASS and disjointness is FAIL.

Literature methods are not automatically experimental arms. Köhl–Hermanns and Fränzle *et al.* are C1 concept comparators because their passive or non-adaptive testing tasks and observation assumptions differ [10], [11]. Hu *et al.* and Miao *et al.* are C2 concept comparators, but their controllable-event authority, objective, observables and stopping rules prevent a fair direct 615A arm without a new mapping decision [8], [9]. F-LAS, IMA and cockpit-display studies are scenario or evaluation context, not algorithm baselines [2], [14], [15]. Confirmatory registration must still freeze the experimental unit, fault construction, equivalence adjudicator, injection confirmation, clock calibration, stochastic repetition/seeds if any, and interval/effect-size method; no literature percentage selects those values [17].

## VI. Results

No effect direction, percentage or superiority is reported.

Every future cell is derived from unique `attemptId` records. Its completion record must identify the claim/RQ, target subset, numerator, denominator, filtering, metric-specific zero-denominator rule, missingness, budget censoring, source fields, unit and charged cost projection. Overlapping flow labels are not summed as an exclusive partition. The synthetic design ledger validates these rules only and is excluded from all result panels.

Coverage metrics retain the finite all-attempt denominator: zero answered attempts among $N > 0$ are reported as $0/N$. Metrics conditional on the answered subset, such as conditional detection error, are N/A when that subset is empty. Truth-dependent cells use explicit truth-scorable subsets: a truth-missing answered or localization attempt remains in coverage, appears in a named missing-truth flow, and leaves only the corresponding truth-scored denominator. Every localization panel first reports decision coverage; truth retention/exclusion and singleton correctness do not silently discard eligible attempts that ended without a decision or lack evaluator truth.

**Placeholder PH-RES-DETECT**
(EXPERIMENT-RESULT)

Separate cells, not one shared denominator: (i) the all-attempt ledger and attempt coverage; (ii) named unconfirmed, invalid, equivalent, undecided, ERROR, abstain and unconcluded flows; (iii) refusal among attempts; (iv) conditional error on the detection answered truth-scorable subset, with truth-scorable coverage and missing-truth flow shown alongside; and (v) miss and false-FAIL on the registered valid detection denominator. Abstain is not PASS/FAIL. Source: EXP-CLTAV-DETECT.

Data are not obtained. This framed panel reserves a single-column result figure. It is not a curve, a mean, or a significance test.

Fig. 3. Reserved detection panel. Not a result.

**Placeholder PH-RES-LOCATE**
(EXPERIMENT-RESULT)

Cells: final $|H|$, true-fault retention/exclusion on the truth-scorable subset, truth-missing flow, decision issued, singleton correctness, refusal and charged resource. Denominator: confirmed in-domain injections, reported separately from the detection answered subset. A valid FAIL with a nonsingleton $H$ is not completed localization. Out-of-domain faults are separate; singleton $h_{\mathrm{normal}}$ is not protocol PASS. Source: EXP-CLTAV-LOCATE.

Data are not obtained. This framed panel reserves a single-column result figure. It is not a curve, a mean, or a significance test.

Fig. 4. Reserved localization panel. Not a result.

TABLE II
Reserved ablation cells. CL-A is computation-only; N/A cells are not silent omissions. FIND-abort waiver is a forbidden control.

| Cell | CL-T | CL-A | CL-TA | CL-LOOP |
|---|---|---|---|---|
| Charged projection | $c_{\mathrm{acq}}$ | $c_{\mathrm{comp}}$ | split | actual |
| No-feedback contrast | yes | N/A | yes | reference |
| Prep/Recover uncharged | if run | N/A | if run | contrast |
| Interval vs point stamp | — | — | — | contrast |
| FIND-abort waiver | forbidden | forbidden | forbidden | forbidden |

### A. CL-RQ2: verdict reliability

### B. CL-RQ2: isolation

### C. CL-RQ3: cost and sensitivity

**Placeholder PH-RES-ABLATION**
(EXPERIMENT-RESULT)

Table II uses the registered ablation factors, not a generic "disable cost accounting." Complete when each named parent metric independently yields a pair on the same registered base scene, seed, construction and contrast, with distinct confirmed resets, disclosed common components and actual cost fields. A missing truth parent does not erase an otherwise eligible cost pair.

Data are not obtained. This box reserves layout. It is not a result.

Recovery may remain a runtime detail, but the evaluator folds it into preparation once and never reports or charges it again as a fifth coordinate. Pre-run review/setup effort is retained in a separate audit total and never charged to an unissued arm action.

## VII. Discussion and Limitations

The first-version domain is a declared single-fault $H_0$, one-step selection, a bound UPLOAD/INFORMATION model, and notation-only architecture views. Out-of-domain faults may remain compatible with an in-domain survivor. A remaining $\{h_{\mathrm{normal}}\}$ is not a protocol PASS. Obtaining ARINC 645 and binding the six distinct triggered obligations closes only their specification contracts; it does not establish any runtime integrity capability.

Engineering work that is not done includes the core estimators, an executable verification engine, Project Configuration and confirmatory runs. Those absences are unfinished work, not method impossibility.

The candidate implementation design is likewise not run admission. Independent mathematical review, an interactive IUT and reset contract, injection/truth confirmation, equivalent-fault adjudication and a justified sample/precision decision remain open. A conforming one-step selector may still show no benefit, more abstention or greater computation on the registered task; those outcomes must remain reportable with coverage and censored cost.

The 23 inspected captures are a selected exploratory corpus with unknown topology, clock qualification, device

configuration, software version and independent root-cause truth. Repeated sessions and shared parser, model or clock components induce dependence and common-error risk. They can exercise intake and reconstruction robustness, but cannot support active counterfactual comparison among the four experimental arms. The bounded reference kernel also abstains rather than treating a frontier limit as incompatibility.

Related work sharpens rather than removes these limits. Passive compatible-run diagnosis does not establish the active protocol lifecycle, while active diagnosis under different control authority does not validate the CL-TAV selector [8]–[10]. An aerospace demonstration has shown that limited observability and model–prototype mismatch can yield wrong localization [13]. In cockpit-display testing, image-derived and simulator-data verdicts share OCL constraints, so high extraction precision is not independent truth for the oracle or product behaviour [15]. The confirmatory design must therefore disclose shared parsers, models, constraints, clocks and truth-generation components and retain discrepancy classes rather than relabel them as product faults.

**Placeholder PH-RES-INTERPRET**
(EXPERIMENT-RESULT)

Discussion of measured gains or losses waits for the reserved result panels. No placeholder may be read as a neutral or negative experimental outcome.

Data are not obtained. This box reserves layout. It is not a result.

Transfer beyond Ethernet laboratory loaders, and any certification argument, remain open (historical RQ6).

## VIII. Conclusion

CL-TAV specifies a charged observation–update–select session for ARINC 615A data loading under declared uncertainty and one resource mode. Its bounded proposition is conditional consistency of interpretation and whole-history update plus a protocol-legal one-step residual-set choice, not universal uncertain-timing diagnosis or global policy optimality. The main-text loop keeps interpretation, compatible update and exclusive stops consistent with the accepted algorithm package. Experimental answers to CL-RQ2 and CL-RQ3 are not available.

**Placeholder PH-CONC-RESULT**
(EXPERIMENT-RESULT)

The concluding experimental sentence stays empty until PH-RES-DETECT, PH-RES-LOCATE and PH-RES-ABLATION are cleared by registered runs.

Data are not obtained. This box reserves layout. It is not a result.

This draft is not submission-ready, not an implemented solver, and not capability establishment.

## REFERENCES


[1] Airlines Electronic Engineering Committee
*ARINC Specification 615A-3: Software Data Loader Using Ethernet Interface*, Airlines Electronic Engineering Committee, 2007, cited by identifier and clause class only; no standard figure is reprinted.

[2] J. Marques, S. Yelisetty, and L. Barros
A framework for loadable airborne systems
*IEEE Aerospace and Electronic Systems Magazine*, vol. 36, no. 5, pp. 38–47, 2021.

[3] J. Tretmans
Conformance testing with labelled transition systems: Implementation relations and test generation
*Computer Networks and ISDN Systems*, vol. 29, no. 1, pp. 49–79, 1996.

[4] A. Petrenko, O. Nguena Timo, and S. Ramesh
Test generation by constraint solving and FSM mutant killing
In *Testing Software and Systems (ICTSS 2016)*, ser. LNCS, vol. 9976, 2016, pp. 36–51.

[5] Y. Jia and M. Harman
An analysis and survey of the development of mutation testing
*IEEE Transactions on Software Engineering*, vol. 37, no. 5, pp. 649–678, 2011.

[6] Z. Yang, R. Huang, C. Cui, N. Niu, and D. Towey
Requirements-based test generation: A comprehensive survey
*ACM Transactions on Software Engineering and Methodology*, 2025.

[7] M. Sampath, S. Lafortune, and D. Teneketzis
Active diagnosis of discrete-event systems
*IEEE Transactions on Automatic Control*, vol. 43, no. 7, pp. 908–929, 1998.

[8] Y. Hu, Z. Ma, and Z. Li
Design of supervisors for active diagnosis in discrete event systems
*IEEE Transactions on Automatic Control*, vol. 65, no. 12, pp. 5159–5172, 2020.

[9] S. Miao, J. Komenda, and A. Lai
Active diagnosis of time-interval automata: Time perspectives
*IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 11 239–11 249, 2025.

[10] M. A. Köhl and H. Hermanns
Model-based diagnosis of real-time systems: Robustness against varying latency, clock drift, and out-of-order observations
*ACM Transactions on Embedded Computing Systems*, vol. 22, no. 4, pp. 68:1–68:48, 2023.

[11] M. Fränzle, T. M. Grosen, K. G. Larsen, and M. Zimmermann
Efficient runtime verification of real-time systems under parametric communication delays
*Formal Aspects of Computing*, vol. 38, no. 2, pp. 17:1–17:32, 2026.

[12] E. Chanthery and Y. Pencolé
Monitoring and active diagnosis for discrete-event systems
*IFAC Proceedings Volumes*, vol. 42, no. 8, pp. 1545–1550, 2009.

[13] E. Chanthery, L. Travé-Massuyès, Y. Pencolé, R. de Ferluc, and B. Dellandrea
Applying active diagnosis to space systems by on-board control procedures
*IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 5, pp. 2568–2580, 2019.

[14] C. Efkemann and J. Peleska
Model-based testing for the second generation of integrated modular avionics
In *2011 IEEE Fourth International Conference on Software Testing, Verification and Validation Workshops*, 2011, pp. 55–62.

[15] H. Sartaj, M. Z. Iqbal, and M. U. Khan
Testing cockpit display systems of aircraft using a model-based approach
*Software and Systems Modeling*, vol. 20, no. 6, pp. 1977–2002, 2021.

[16] B. J. M. Grün, D. Schuler, and A. Zeller
The impact of equivalent mutants
In *2009 International Conference on Software Testing, Verification, and Validation Workshops*, 2009, pp. 192–199.

[17] A. Arcuri and L. Briand
A practical guide for using statistical tests to assess randomized algorithms in software engineering
In *Proceedings of the 33rd International Conference on Software Engineering*, 2011, pp. 1–10.

[18] Joint Committee for Guides in Metrology
Evaluation of measurement data — guide to the expression of uncertainty in measurement
JCGM, Tech. Rep. JCGM 100:2008, 2008.

[19] Joint Committee for Guides in Metrology
Evaluation of measurement data — the role of measurement uncertainty in conformity assessment
JCGM, Tech. Rep. JCGM 106:2012, 2012.

[20] Airlines Electronic Engineering Committee
*ARINC Specification 645-1: Common Terminology and Functions for Software Distribution and Loading*, Airlines Electronic Engineering Committee, 2021, 615A-triggered SOURCE/SEMANTIC bind; capabilities not established.